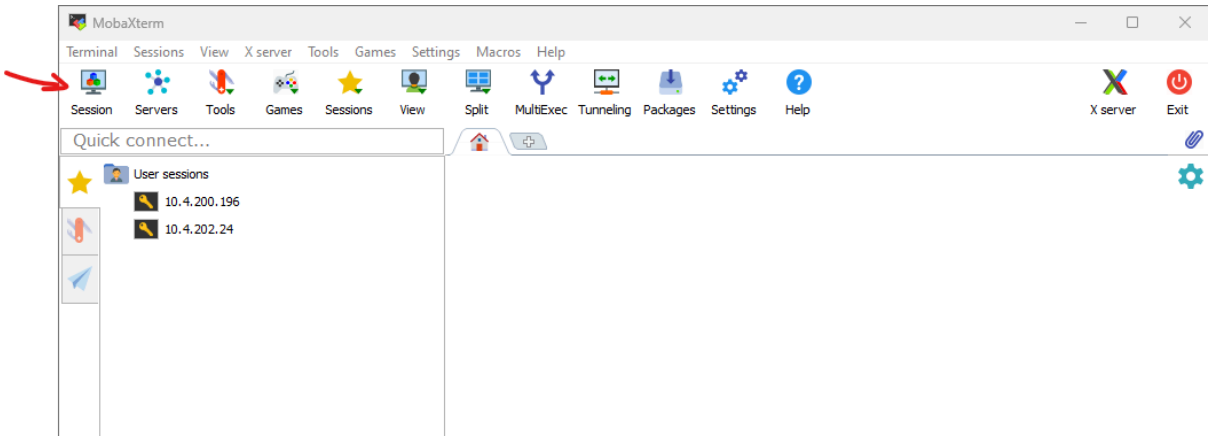
Previamente encendemos nuestra máquina virtual con Ubuntu-Server y sacamos la IP utilizando el comando **ifconfig**

Ubuntu-Server [Corriendo] - Oracle VM VirtualBox

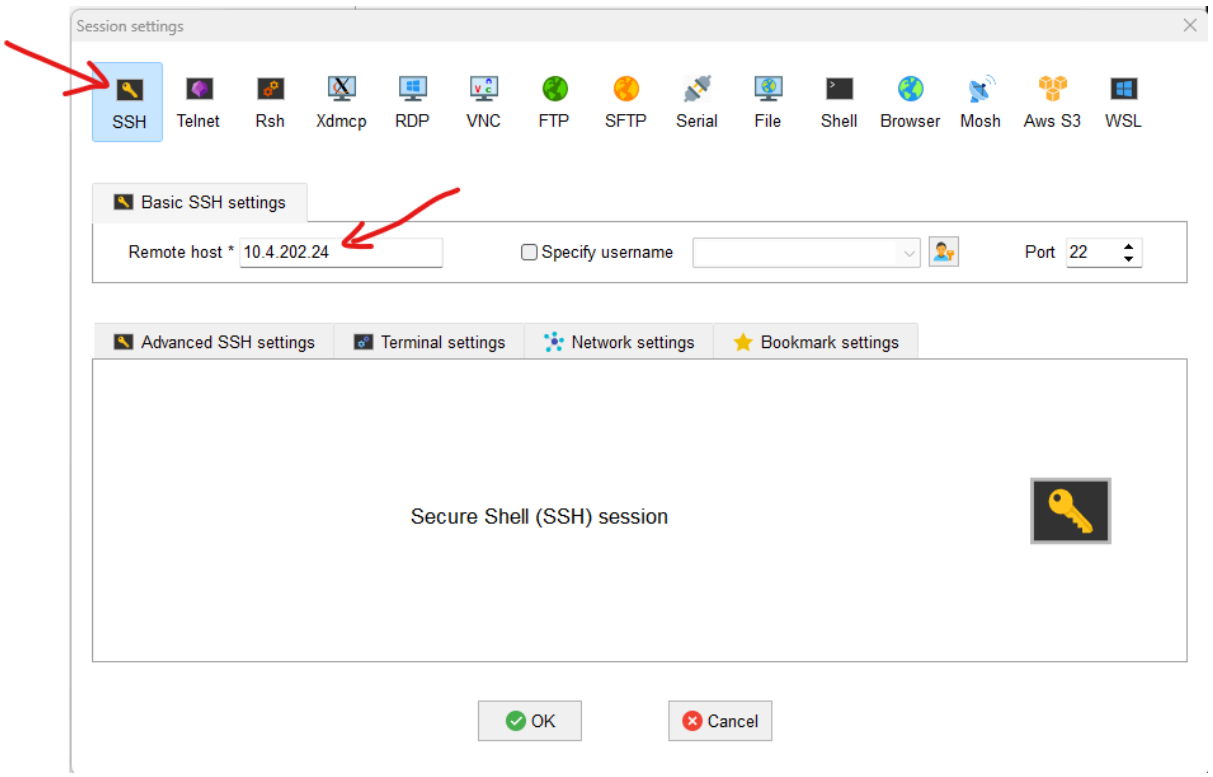
```
  System load:  0.0                Processes:               103
  Usage of /:   32.7% of 8.02GB    Users logged in:         0
  Memory usage: 23%                IPv4 address for enp0s3: 10.4.202.24
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

smr@smr-server:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether 02:42:28:71:bc:c6  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.4.202.24  netmask 255.255.252.0  broadcast 10.4.203.255
        inet6 fe80::a00:27ff:fe56:3a4f  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:56:3a:4f  txqueuelen 1000  (Ethernet)
        RX packets 1415  bytes 140530 (140.5 KB)
        RX errors 0  dropped 34  overruns 0  frame 0
        TX packets 49  bytes 9106 (9.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 72  bytes 5636 (5.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 72  bytes 5636 (5.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Instalamos MobaXTerm.
Ejecutamos el programa

Le damos a **Session**



Seleccionamos **SSH** y metemos la dirección IP del servidor de Ubuntu. En este caso **10.4.202.24.**



Una vez conectamos nos pedirá usuario y contraseña.

```
3. 10.4.202.24

                • MobaXterm Personal Edition v24.2 •
              (SSH client, X server and network tools)

 ▸ SSH session to smr@10.4.202.24
   • Direct SSH      :  ✔
   • SSH compression :  ✔
   • SSH-browser     :  ✔
   • X11-forwarding  :  ✔  (remote display is forwarded through SSH)

 ▸ For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-45-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Oct 15 05:57:47 PM UTC 2024

  System load:  0.0               Processes:               112
  Usage of /:   32.7% of 8.02GB   Users logged in:         1
  Memory usage: 26%               IPv4 address for enp0s3: 10.4.202.24
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Oct 15 16:41:45 2024 from 10.4.201.248
smr@smr-server:~$
```

Queremos automatizar un script con **cron.** Para ello vamos a generar primero nuestro script.
Por ejemplo vamos a crear el archivo **copia-conf.sh** dentro de la carpeta **/home/smr/**

```
2. 192.168.18.42
  GNU nano 7.2                                  copia-conf.sh *
#!/bin/bash
cp /etc/apache2/sites-available/smr*.conf /home/smr/Documentos/
tar cvf /home/smr/Documentos/$(date +"%Y%m%d")_apache_config.tar.gz /home/smr/Documentos/*.conf
rm /home/smr/Documentos/*.conf
```

Esto lo que va a hacer es copiar los archivos **smr*.conf** dentro de una carpeta, luego comprimir esos archivos dentro de un **tar.gz** asignando de nombre **20241016_apache_config.tar.gz** y finalmente eliminando los archivos ***.conf** sobrantes

Ahora le damos permisos de ejecución con el comando **chmod +x copia-conf.sh**

```
root@smr-server:/home/smr# chmod +x copia-conf.sh
root@smr-server:/home/smr# ls
copia-conf.sh  Documentos  index.html  lista_comandos  smr.martinez.conf
root@smr-server:/home/smr# ./copia-conf.sh
tar: Removing leading `/' from member names
```

Para comprobar que funciona usamos **./copia-conf.sh**
Vemos que efectivamente funciona el script.

```
root@smr-server:/home/smr/Documentos# ls
20241015_apache_config.tar.gz
root@smr-server:/home/smr/Documentos#
```

Ahora procedemos a editar **crontab -e** para que se ejecute el script de forma automática.
Para ello usamos el comando **crontab -e**. Seleccionamos **1. /bin/nano**

```
smr@smr-server:~$ crontab -e
no crontab for smr - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]:
```

Dentro de **crontab** asignamos cuando queremos que se ejecute el script.
En este caso * * * * * (Cada minuto) y le decimos el script que tiene que ejecutar, **/home/smr/copia-conf.sh**

```
5. 10.4.202.24
  GNU nano 7.2                /tmp/crontab.H85e99/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
* * * * * /home/smr/copia-conf.sh
```

Ya tenemos nuestro script programado con crontab para que se ejecute cada minuto como lo hemos programado.

Pero como el ejercicio nos pide que sea **TODOS LOS DIAS A LAS 19:40** lo tenemos que cambiar a esto en **contrab -e**

```
2. 10.4.202.116
  GNU nano 7.2                    /tmp/crontab.B86ZCB/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
40 19 * * * /home/smr/copia-conf.sh
```